CHETONA

A SECURE AWARENESS LANGUAGE

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1. Introduction

CHETONA (Bengali for "Consciousness") is a domain-specific language (DSL) designed for logical execution with a focus on human-readable keywords. Unlike traditional languages that use Western syntax, CHETONA leverages the rich vocabulary of the Bengali language to create a bridge between natural human thought and machine logic.

The compiler for CHETONA is implemented in C++17, utilizing a robust pipeline of Lexical Analysis, Parsing (AST construction), and Interpretation to provide a seamless development experience.

2. Language Overview

File Extension: `.chn`

Syntax Philosophy: Every keyword in CHETONA is carefully chosen to represent its logical operation in a way that feels intuitive to native speakers while maintaining technical precision.

3. Function Table (30 Unique Functions)

The following table lists the core functions of the CHETONA language, mapping standard operations to their unique Bengali keywords.

Standard Function	CHETONA Keyword	Short Description
Print Output	`prodorshon()`	Outputs text or value to console
Input Value	`grohon()`	Reads user input from console
Add	`shonkolon()`	Adds two numeric values
Subtract	`biyojon()`	Subtracts one value from another
Multiply	`gunon()`	Multiplies two numeric values
Divide	`vabhon()`	Divides one value by another
Modulo	`vagshesh()`	Finds the remainder of division
Compare	`tulona()`	Compares two values for logic
Equals	`shomota()`	Checks if two values are equal

Standard Function	CHETONA Keyword	Short Description
Greater Than	`adhik()`	Checks if first is larger
Less Than	`kolpo()`	Checks if first is smaller
Boolean True	`shotto`	Logical true constant
Boolean False	`mithya`	Logical false constant
Conditional (If)	`shorto`	Initializes an 'if' block
Else	`onno-thay`	Initializes an 'else' block
While Loop	`ghurnon()`	Executes while condition is true
Break	`chhed()`	Exits the current loop
Continue	`punoraoy()`	Skips to next loop iteration
Return	`protyaborton()`	Returns value from a block
Open File	`udghaton()`	Opens a file for operations
Close File	`shongbidhan()`	Closes an open file stream
Read File	`pathon()`	Reads content from a file
Write File	`likhon()`	Writes content to a file
Start Program	`aromvho()`	The main entry point function

Standard Function	CHETONA Keyword	Short Description
End Program	`shomapti()`	Terminates program execution
Convert to String	`borno()`	Casts value to string type
Convert to Number	`onko()`	Casts value to number type
System Time	`kal()`	Gets the current system time
Random Value	`accirik()`	Generates a random integer
Exit Compiler	`biday()`	Closes the CHETONA shell

4. Sample CHETONA Code

```
// CHETONA Demonstration Program
aromvho() {
    prodorshon("Welcome to CHETONA!");

    var x = grohon(); // Get input
    var y = 10;

    shorto (shomota(x, y)) {
        prodorshon("Values are equal.");
    } onno-thay {
        prodorshon("Values are different.");
    }

    shomapti();
}
```

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